

Causality and Machine Learning: *Causal Discovery*

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- 1 Causal discovery
- 2 Learning from observational data
 - Constraint-based approach
 - Score-based approach
- 3 Learning beyond Markov equivalence classes
 - Learning from interventional data
 - Learning with functional assumptions
- 4 Evaluation

1. Causal discovery

Causal Discovery

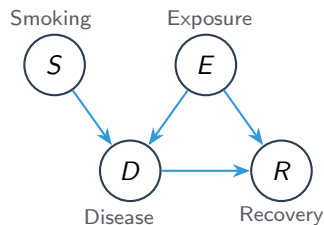
- So far, we have assumed that we are given a causal model
- But where do the models come from?

Where do the causal models come from?

- An expert can construct them
- We can conduct randomized controlled trials
- We can learn from data

Expert construction

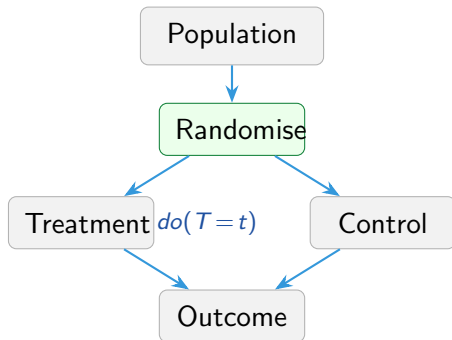
- Pros: Incorporates domain knowledge, no data needed
- Cons: Works only for small and simple models, requires domain knowledge, difficult to quantify uncertainty



A domain expert encodes their mechanistic understanding directly into a DAG.

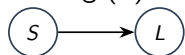
Randomised controlled trials

- Pros: Gold standard, eliminates confounding by design, identifies causal effect directly
- Cons: Often expensive, unethical or impossible, only tests pre-defined hypotheses, does not discover full structure

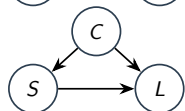


Example: Randomised controlled trial in causal discovery

Smoking (S) and lung cancer (L) are correlated. What are possible causal structures?



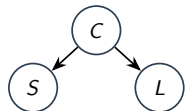
Causal relation, no confounding



Causal relation, confounding



No causation, no confounding

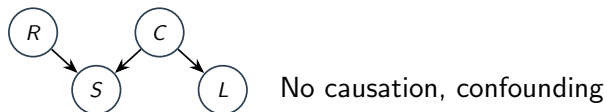
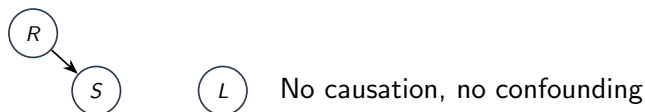
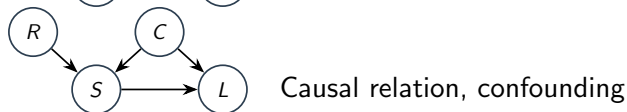
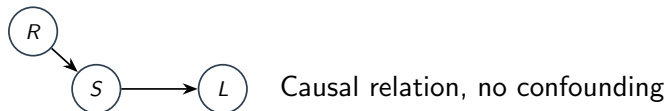


No causation, confounding

- Assumptions: 1) S precedes L, 2) No feedback cycles, 3) Multiple confounders are equivalent to a single confounder (C)

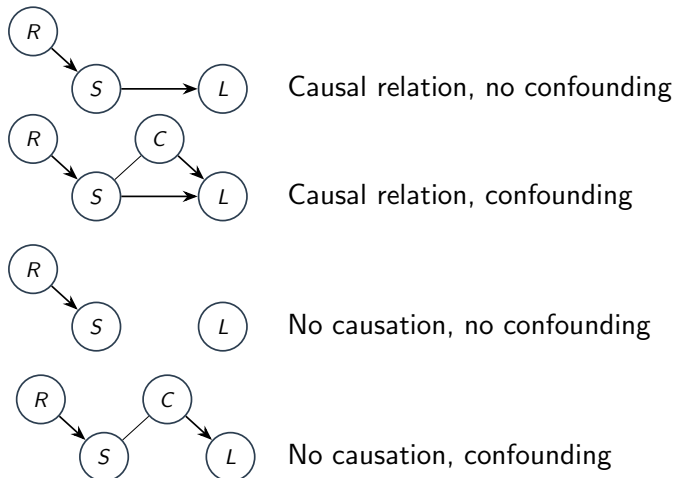
Example: Randomised controlled trial in causal discovery

- How to decide which structure is correct?
- Use a randomized controlled trial



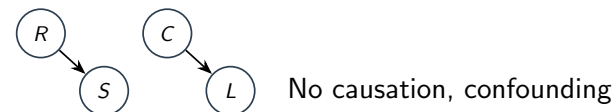
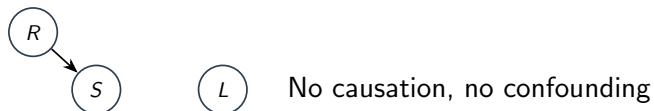
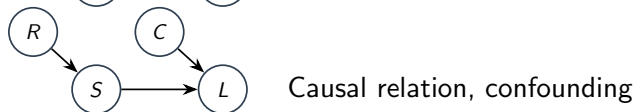
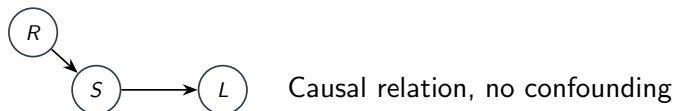
Example: Randomised controlled trial in causal discovery

- Randomization removes dashed edges

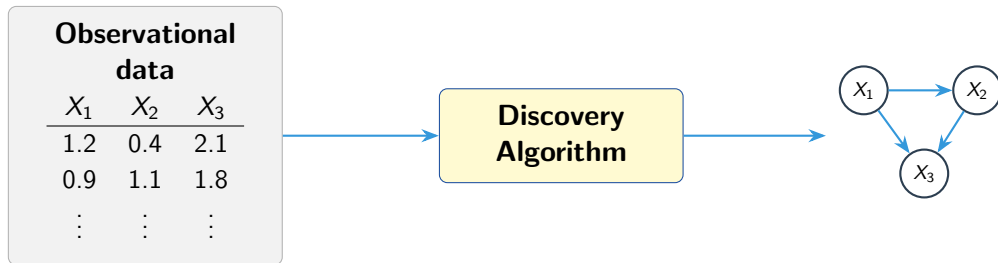


Example: Randomised controlled trial in causal discovery

- Thus, if association persists, S is a cause of L



Learning from data - Causal Discovery



- Pros:

- Scales to many variables
- No expert in the loop required
- Applicable when interventions are impossible

- Cons:

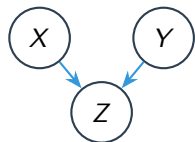
- Requires strong assumptions
- Identifiability is limited (equivalence classes)
- Sensitive to violations of assumptions
- Computationally challenging at scale

Subtasks of causal discovery

- Structure learning
 - Which variables are causally related? What are the edges?
 - Output: $\tilde{\mathcal{G}}$: estimated DAG or CPDAG over observable variables
- Parameter learning
 - What are the strengths / functions of the causal relationships?
 - Output: $\mathbf{F}$, $P(\mathbf{U})$
- We focus on structure learning

Learning from data

Observational data

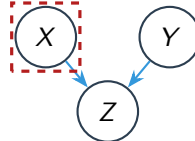


passive observation

- Pros: No intervention needed; cheap; abundant
- Cons: Limited identifiability; confounders problematic

Interventional data

$do(X=x)$



active manipulation

- Pros: Breaks Markov equivalence; more identifiable
- Cons: Costly; sometimes unethical; not always possible

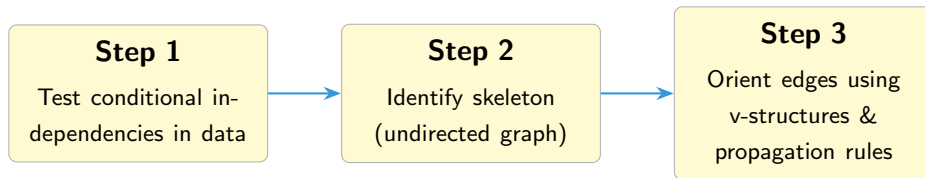
2. Learning from observational data

Learning from observational data

- Basic idea:
 - Assume there is an unknown causal model which generated data.
 - The causal model implies that the observed data have some specific properties
 - Learning: Observe these properties and go backwards to figure out what was the model.
- Two main approaches:
 - Constraint-based
 - Score-based

2.1. Constraint-based approach

Constraint-based approach: High-level idea



Key insight: Conditional independencies in the data constrain which DAGs are consistent with the data.

$$X \perp\!\!\!\perp Y \mid Z \Leftrightarrow \text{d-sep}_{\mathcal{G}}(X; Y \mid Z);$$

Conditional independence — refresher

Definition

X and Y are **conditionally independent given** Z , written $X \perp\!\!\!\perp Y \mid Z$, if

$$P(X \mid Y, Z) = P(X \mid Z) \iff P(X, Y \mid Z) = P(X \mid Z) P(Y \mid Z).$$

Three cases:



Structure	$X \perp\!\!\!\perp Y?$	$X \perp\!\!\!\perp Y \mid Z?$
Chain	No	Yes
Fork	No	Yes
Collider	Yes	No

How to test if two variables are conditionally independent?

Testing conditional independence: χ^2 test (example)

Setting: Discrete variables. Test $X \perp\!\!\!\perp Y \mid Z$.

Procedure

- 1 For each value z of Z , form the contingency table of (X, Y)
- 2 Compute observed counts $O_{ij|z}$ and expected under independence:

$$E_{ij|z} = \frac{O_{i\cdot|z} \cdot O_{\cdot j|z}}{n_z}$$

- 3 Sum the χ^2 statistics over strata z :

$$\chi^2 = \sum_z \sum_{i,j} \frac{(O_{ij|z} - E_{ij|z})^2}{E_{ij|z}}$$

- 4 Compare to $\chi^2_{(r-1)(c-1) \cdot |Z|}$ distribution

Toy example:

X : Smoker (Y/N), Y : Cancer (Y/N), Z : Age group

	$Z = \text{young}$	
	Cancer Y	Cancer N
Smoker Y	10	40
Smoker N	5	45

$p\text{-value} < 0.05$: reject $X \perp\!\!\!\perp Y \mid Z$

Continuous data: use partial correlation or kernel tests (HSIC).

Challenges with conditional independence testing

- Statistical issues:
 - Finite sample size: asymptotic tests are unreliable for small n
 - Large conditioning sets: Statistical power decreases when conditioning sets become larger
 - Discrete vs continuous variables: Different tests needed
 - Multiple testing: Potentially exponentially many tests

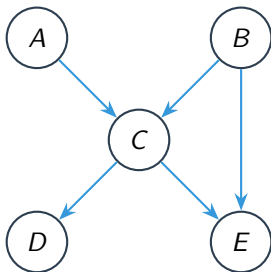
d -separation revisited

Definition

A path π between X and Y is **blocked** by a set Z if either: (i) π contains a chain $A \rightarrow B \rightarrow C$ or fork $A \leftarrow B \rightarrow C$ where $B \in Z$, or (ii) π contains a collider $A \rightarrow B \leftarrow C$ where $B \notin Z$ and no descendant of B is in Z .

X and Y are **d-separated** by Z if *all* paths between them are blocked by Z .

Example:



- $d\text{-sep}_{\mathcal{G}}(A; B \mid \emptyset)$?
- $d\text{-sep}_{\mathcal{G}}(A; B \mid \{C\})$?
- $d\text{-sep}_{\mathcal{G}}(A; E \mid \{C\})$?
- $d\text{-sep}_{\mathcal{G}}(A; E \mid \{C, B\})$?
- $d\text{-sep}_{\mathcal{G}}(D; E \mid \{C\})$?

Conditional independencies and DAGs

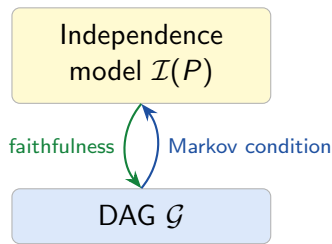
Markov condition:

$$X \perp\!\!\!\perp Y \mid Z \iff \text{d-sep}_{\mathcal{G}}(X; Y \mid Z)$$

Faithfulness:

$$X \perp\!\!\!\perp Y \mid Z \implies \text{d-sep}_{\mathcal{G}}(X; Y \mid Z)$$

Under **both** assumptions, the set of conditional independencies in the distribution is *exactly* the set of d-separations in the DAG. This allows us to go from data to graph structure.



only up to Markov equivalence!

Exercise

We have conducted conditional independence tests and got the following results. Construct a DAG that expresses these conditional independencies and dependencies exactly.

Graph 1	Graph 2
$V = \{A, B, C, D\}$	$V = \{X, Y, Z, W\}$
$A \perp B$	$W \perp X$
$B \perp D \mid \{A\}$	$X \perp Z$
$A \not\perp C$	$W \perp Z$
$A \perp C \mid \{D\}$	

Constraint-based approach: Assumptions

- **Causal Markov condition:**

- Every variable is independent of its non-effects (non-descendants) given its direct causes (parents).
- *Consequence:* every d-separation implies a conditional independence in P .

- **Faithfulness:**

- No conditional independence in P that is *not* entailed by d-separation in $\mathcal{G}$.
- *Consequence:* the independence model of P exactly matches d-separations of $\mathcal{G}$.

- **Causal sufficiency:**

- All common causes of observed variables are also observed (no hidden confounders).
- *Consequence:* the true DAG is over the observed variables.

- **Acyclicity:** The causal graph is as DAG (no feedback loops)

- **Conditional independence oracle:** Conditional independence tests are errorless (equivalent to assuming infinite data and no selection bias)

What is the PC algorithm?

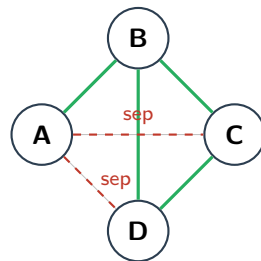
Goal

Learn the **Markov equivalence class** of a DAG from observational data using *conditional independence tests*.

Two main phases:

- 1 **Skeleton learning** — find which pairs of variables are *adjacent* (connected by an edge)
- 2 **Orientation** — determine edge directions where possible

Named after its inventors: **P**eter Spirtes & **C**larke Glymour (1991).



Skeleton after Phase 1

Skeleton

Definition

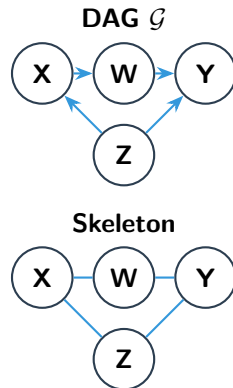
The **skeleton** of a DAG $\mathcal{G}$ is the *undirected* graph obtained by replacing every directed edge $X \rightarrow Y$ with an undirected edge $X - Y$.

Key insight

Two variables X, Y are **non-adjacent** in the true DAG if and only if there exists a set $\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Y\}$ such that

$$X \perp\!\!\!\perp Y \mid \mathbf{S}$$

This set $\mathbf{S}$ is called the **separating set** $\text{Sep}(X, Y)$.



The PC skeleton algorithm: overview

Algorithm sketch

- ① Start with the **complete undirected graph** $\mathcal{C}$ on p variables.
- ② For $\ell = 0, 1, 2, \dots$:
 - For each adjacent pair (X, Y) in the current graph, look for a **separating set** of size ℓ among the *neighbours* of X (or Y).
 - If $X \perp\!\!\!\perp Y \mid \mathbf{S}$ for some $|\mathbf{S}| = \ell$: remove edge $X - Y$ and store $\mathbf{S}$ as $\text{Sep}(X, Y)$.
- ③ Stop when every adjacent pair (X, Y) has $|\text{adj}(X) \setminus \{Y\}| < \ell$.

Why start small?

Testing subsets of increasing size avoids redundant large-set tests. Once an edge is removed, the number of neighbours shrinks — later tests are cheaper.

Complexity advantage:

In sparse graphs, edges are removed early with small conditioning sets, so the neighbourhood never grows large. Worst case is still exponential, but typical cases are tractable.

Why condition on *neighbours*, not all variables?

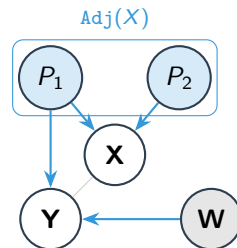
Markov Condition

Every variable X is conditionally independent of its *non-descendants* given its *parents*:

$$X \perp\!\!\!\perp \text{NonDe}(X) \mid \text{Pa}(X)$$

Consequence

We *never* need to condition on variables outside the neighbourhood of X (or Y) to find a separating set. This limits the search space dramatically.



$$\text{Sep}(X, Y) \subseteq \{P_1, P_2\}$$

(no need to include W)

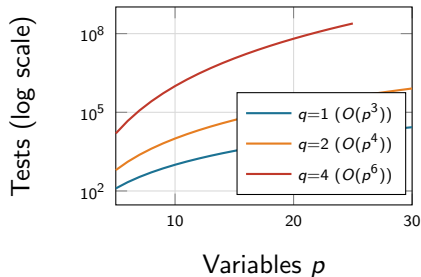
Complexity and practical considerations

Worst-case complexity

$O(p^{q+2})$ conditional independence tests, where q is the maximum *neighbourhood size* in the true graph.

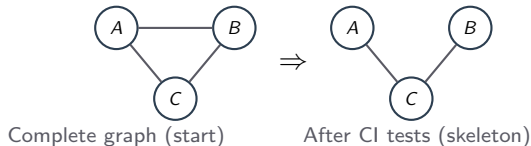
In sparse graphs (q small): near-polynomial — *scales to hundreds of variables*.

In dense graphs ($q \approx p$): exponential — unavoidable in worst case.

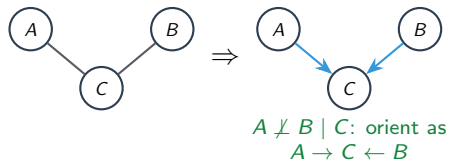


PC algorithm

Phase 1: Skeleton discovery



Phase 2: V-structure orientation

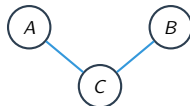


Algorithm (sketch)

- 1 Start with complete undirected graph
- 2 For $k = 0, 1, 2, \dots$:
 For each adjacent pair (X, Y) :
 Test $X \perp\!\!\!\perp Y \mid S$ for all
 $S \subseteq \text{adj}(X) \setminus \{Y\}$, $|S| = k$
 If independent: remove edge,
 store sep set S_{XY}
- 3 Orient v-structures:
 $X - Z - Y$ with $Z \notin S_{XY}$
 $\Rightarrow X \rightarrow Z \leftarrow Y$
- 4 Apply Meek rules to orient remaining edges

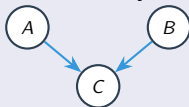
V-structure orientation

Given a triple of nodes A , B , C with edges $A - C$ and $B - C$ and no edge between A and B (unshielded collider), how to determine whether it is an immorality or not?



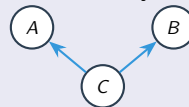
Case 1: Immorality

C cannot be in any $Sep(A, B)$



Case 2: No immorality

C must be in every $Sep(A, B)$



To decide whether there is an immorality or not, it suffices to look at the separating set.

Meek rules for orienting arcs

Given a **completed partially directed acyclic graph (CPDAG)**,
Meek's rules propagate orientation constraints without introducing new v-structures.

Context: After running a skeleton-finding algorithm (e.g. PC), we have:

- Some edges already **oriented** (from v-structure detection)
- Some edges still **undirected**

Goal: Orient as many remaining undirected edges as possible while remaining consistent with:

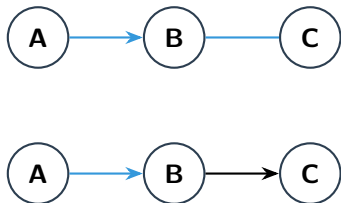
- Acyclicity (no directed cycles)
- No new v-structures

Meek rules: R1 and R2

Rule R1 — away from collider

If $A \rightarrow B \text{ --- } C$ and $A \not\rightarrow C$, orient $B \rightarrow C$.

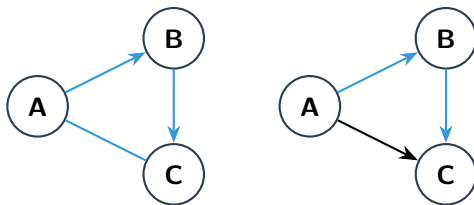
Reason: orienting $C \rightarrow B$ would create a new collider at B.



Rule R2 — away from cycle

If $A \rightarrow C$, $A \rightarrow B \rightarrow C$, and $A \text{ --- } C$, orient $A \rightarrow C$.

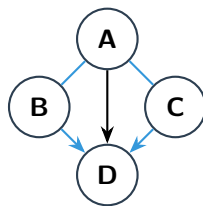
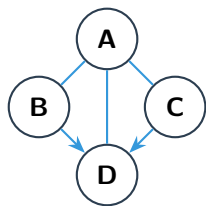
Reason: orienting $C \rightarrow A$ would create a directed cycle.



Meek rules: R3 and R4

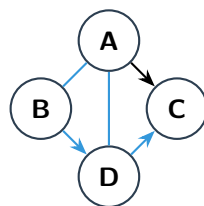
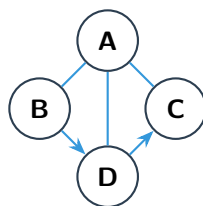
Rule R3 — Avoid new v-structure

If $A - B$, $A - C$, $A - C$, $D \rightarrow B$, $C \rightarrow B$, orient $A \rightarrow B$



Rule R4 — Avoid cycles

If $A - B \rightarrow C \rightarrow D$ and $A - D$, $A - C$, orient $A \rightarrow D$.



Meek rules

Key Theorem (Meek, 1995):

Applying R1–R4 exhaustively to a CPDAG yields a *maximally oriented* PDAG — no further orientations can be inferred without making additional assumptions.

Important

Meek rules are **sound and complete**: if an edge *can* be oriented consistently within the Markov equivalence class, the rules *will* orient it.

Exercise: PC algorithm

Learn a CPDAG given a conditional independence oracle. A jupyter notebook with the oracle can be found at the course webpage.

- <https://shorturl.at/efTNq>
- <https://johandh2o.github.io/causal-inference-summer-school>



Correctness of the PC Algorithm

Theorem (Spirtes, Glymour, Scheines 1993)

Assume causal Markov, faithfulness, causal sufficiency, and acyclicity.

*Given the oracle conditional independence relations (perfect tests), the PC algorithm returns the **correct CPDAG** — the unique graphical representation of the Markov equivalence class of the true DAG.*

What this means

- All v-structures are correctly identified
- All edges that *can* be oriented consistently with the data *are* oriented
- Remaining undirected edges represent genuine Markov equivalence (both orientations are consistent)

Finite sample caveat

With real tests (not oracle), consistency holds only asymptotically.

For finite n : errors in CI tests $\Rightarrow$ errors in the output graph.

Identifiability up to Markov Equivalence

Two DAGs are **Markov equivalent** if they encode the same conditional independencies.



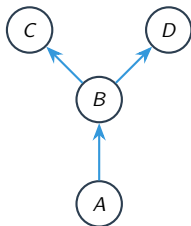
Bottom line

From observational data alone (under the four assumptions), the **best we can do** is recover the CPDAG — not the true DAG.

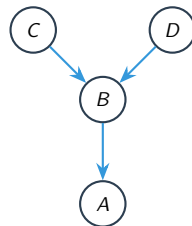
Identifiability example

The degree of identifiability depends heavily on the structure.

None of the directions identifiable!



All directions identifiable!



How realistic are the assumptions?

- Four assumptions about the data-generating process
 - Causal Markov condition
 - Faithfulness
 - Causal sufficiency
 - Acyclicity
- One assumption about access to the data
 - Conditional independence oracle

Causal Markov condition

Formal Statement

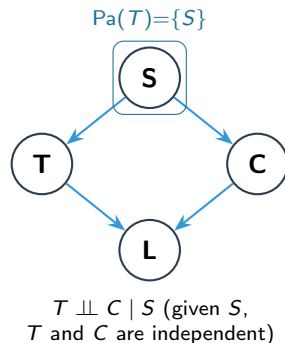
Let $\mathcal{G}$ be the true causal DAG and P the joint distribution. The **Causal Markov Condition** states:

$$X \perp\!\!\!\perp_P \text{NonDe}(X) \mid \text{Pa}(X) \quad \forall X$$

Equivalent formulations:

- *Factorization*: $P(\mathbf{V}) = \prod_i P(X_i \mid \text{Pa}(X_i))$
- *Global Markov*: d -separation in $\mathcal{G}$ implies CI in P

How realistic? Very plausible if $\mathcal{G}$ correctly represents all direct causes. It follows from standard probabilistic/structural equation model assumptions.



Faithfulness

Formal Statement

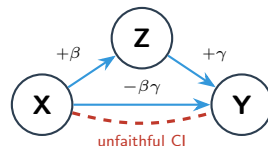
P is **faithful** to $\mathcal{G}$ if every conditional independence in P corresponds to a d -separation in $\mathcal{G}$:

$$X \perp\!\!\!\perp_P Y \mid \mathbf{S} \Rightarrow \text{d-sep}_{\mathcal{G}}(X; Y \mid \mathbf{S})$$

What it rules out: cancellations of paths where two paths carry exactly equal-and-opposite effects, creating a spurious independence.

Violation example

$X \rightarrow Z \rightarrow Y$ and $X \rightarrow Y$ with perfectly cancelling effects: $X \perp\!\!\!\perp Y$ marginally, but X and Y are adjacent in the true graph.



Total effect $X \rightarrow Y$:
 $\beta\gamma + (-\beta\gamma) = 0$

$\Rightarrow$ **unfaithful:** $X \perp\!\!\!\perp Y$
 but X - Y edge exists!

How realistic is faithfulness?

Arguments in favour:

- Exact cancellations require *measure-zero* parameter settings — “almost all” distributions over a DAG are faithful.
- Generic models (e.g. linear Gaussian with generic coefficients) are faithful with probability 1.

Arguments against:

- In *finite samples*, near-unfaithfulness causes low power — CI tests may fail to reject independence even when the relationship is real.
- Deterministic relationships (logical constraints) violate faithfulness.

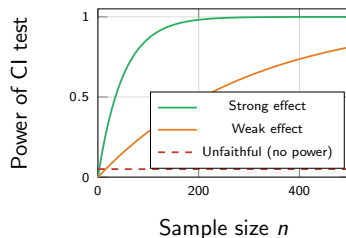
Violations of faithfulness: remedies

Near-faithfulness in finite samples:

- Even if parameters are not exactly cancelling, with small n , the power of CI tests may be insufficient to detect weak associations.
- PC may incorrectly *remove edges* that carry small but real effects.

Conservative alternatives:

- **Conservative PC (CPC)**: only orient v-structures when all subsets agree.
- **Majority-rule PC**: requires a majority of conditioning sets to agree on independence.



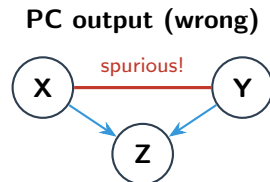
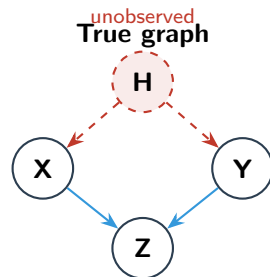
Causal sufficiency

Formal Statement

The set of observed variables $\mathbf{V}$ is **causally sufficient** if for every pair $X, Y \in \mathbf{V}$, all their *common causes* are also in $\mathbf{V}$.

No hidden/latent common causes (confounders). Implications:

- Without sufficiency, marginal dependencies remain after conditioning on all observed variables.
- PC will erroneously include edges between variables that are only related through hidden causes.



How realistic is causal sufficiency?

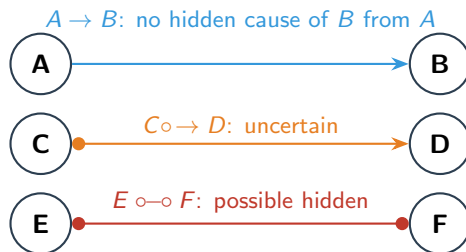
Often violated in practice:

- **Omitted variables:** we never measure everything.
- **Gene expression:** unmeasured transcription factors confound observed gene–gene associations.
- **Econometrics:** unmeasured “ability” confounds education–income links.
- **Medicine:** unmeasured lifestyle factors confound treatment–outcome relations.

Remedy: FCI Algorithm

The **Fast Causal Inference (FCI)** algorithm relaxes causal sufficiency. It outputs a **PAG** (Partial Ancestral Graph) that represents uncertainty about hidden confounders.

PAG Edge Marks



Acyclicity

DAG Assumption

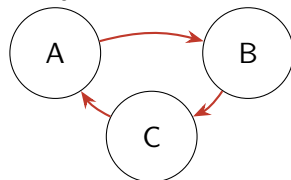
PC assumes the true causal structure is a **Directed Acyclic Graph** — no feedback cycles among the measured variables at the relevant time scale.

Violations occur when:

- Variables represent aggregated time-series (e.g. yearly averages masking feedback).
- Biological regulation (gene regulatory networks often have cycles).
- Economic equilibria with simultaneous determination.

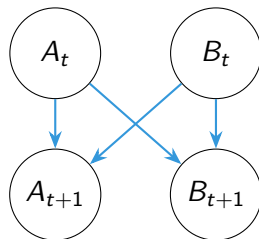
Remedies: Time-series methods, cyclic causal models.

Cycle: not a DAG



$A \rightarrow B \rightarrow C \rightarrow A$: cycle

Time-series workaround



Acyclic across time

Errors in CI Tests

Assumption: Correct CI Tests

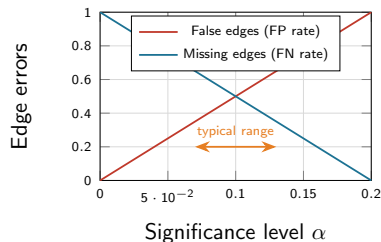
PC is derived for the *oracle* setting (infinite data, exact CIs). In practice:

- We use statistical tests
- All tests have *Type I* and *Type II* errors.
- **Errors compound:** the skeleton is the foundation for orientation — one wrong edge can cascade.

Significance level α :

Larger $\alpha \rightarrow$ more edges (reject null hypothesis often, denser graph, more false edges).

Smaller $\alpha \rightarrow$ fewer edges (reject null hypothesis rarely, sparser graph, more false non-edges).



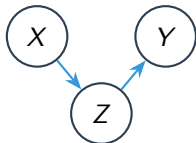
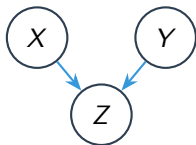
What happens when test results are inconsistent?

CI tests can produce results that are **logically inconsistent** — no single DAG satisfies all of them simultaneously.

Example: Tests say: $X \not\perp Z$, $Y \not\perp Z$, $X \perp Y$ (collider) but also $X \perp Y \mid Z$ (not a collider).

There does not exist a DAG faithful to the tests.

Which graph should we choose?



Common causes

- Type I / II errors in CI tests
- Different conditioning sets giving conflicting results
- Faithfulness violations
- Model misspecification

Practical responses

- Use majority vote over sep sets
- Conservative: keep edge if *any* test retains it
- Report uncertainty; return PAG (partial ancestral graph) instead
- Weight test results by confidence

What happens when assumptions are violated?

Assumption	If violated. . .	Consequence for PC
Causal Markov	Distribution is not Markov to the graph	CI tests inconsistent; wrong graph
Faithfulness	Exact/near path cancellation	Missing edges (false absences); wrong v-structures
Causal Sufficiency	Hidden common causes	Spurious edges added between unconnected variables
Acyclicity	Feedback loops in system	Orientation rules break down; inconsistent orientations
Correct CI tests	Finite sample, non-Gaussian	Type I/II errors propagate; skeleton errors compound

2.2. Score-based approach

Score-based structure learning

Core idea:

- 1 Define a *score function* $S(\mathcal{G}, \mathcal{D})$ that measures how well a DAG $\mathcal{G}$ fits data $\mathcal{D}$
- 2 Search over the space of DAGs (or CPDAGs) to maximize the score

Contrast with constraint-based:

- PC tests *conditional independencies*
- Score-based methods *optimize* a global objective
- Typically more robust to small samples

$$\hat{\mathcal{G}} = \arg \max_{\mathcal{G} \in \text{DAGs}} S(\mathcal{G}, \mathcal{D}) \quad \text{subject to: } \mathcal{G} \text{ is a DAG (no directed cycles)}$$

Where do scores come from? — Likelihood

Log-likelihood score:

$$S_{LL}(\mathcal{G}, \mathcal{D}) = \log P(\mathcal{D} \mid \hat{\theta}_{\mathcal{G}}, \mathcal{G})$$

where $\hat{\theta}_{\mathcal{G}}$ are the MLE parameters.

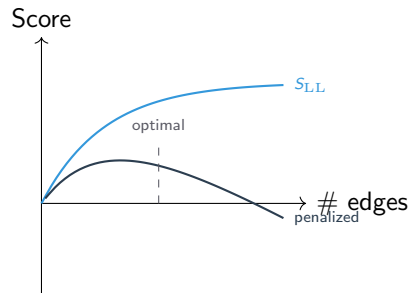
Using the Markov factorization:

$$\log P(\mathcal{D} \mid \mathcal{G}) = \sum_{i=1}^n \sum_{j=1}^N \log P(X_i^{(j)} \mid \text{Pa}_i^{(j)})$$

Problem: Overfitting

Adding edges *always* increases S_{LL} .
A complete graph wins trivially.

Solution: penalize complexity.



Complexity penalty balances fit vs. parsimony (Occam's razor).

Common score functions

BIC — Bayesian Information Criterion

$$\text{BIC}(\mathcal{G}) = \log P(\mathcal{D} \mid \hat{\theta}, \mathcal{G}) - \frac{d_{\mathcal{G}}}{2} \log N$$

- $d_{\mathcal{G}}$: # free parameters in $\mathcal{G}$
- N : sample size

BDeu / BGe — Bayesian Score

$$S_{\text{Bayes}}(\mathcal{G}) = \log P(\mathcal{G}) + \log P(\mathcal{D} \mid \mathcal{G})$$

- $P(\mathcal{D} \mid \mathcal{G})$ marginalises out θ
- BDeu: discrete variables (Dirichlet prior)
- BGe: continuous / Gaussian variables

Properties

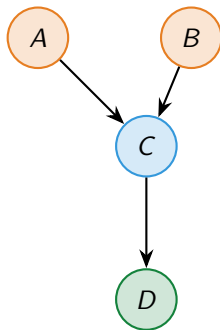
- Consistency: recovers true DAG as $N \rightarrow \infty$
- Score equivalence: Markov equivalent DAGs get equal scores

$$\text{Score decomposability: } S(\mathcal{G}) = \sum_{i=1}^n s_i(X_i, \text{Pa}_i)$$

Each variable's local score depends only on its parents $\Rightarrow$ enables efficient search!

Score decomposability — Illustration

Example DAG: $A \rightarrow C$, $B \rightarrow C$,
 $C \rightarrow D$



Score factorises as:

$$S(\mathcal{G}) = \underbrace{s(A, \emptyset)}_{\text{no parents}} + \underbrace{s(B, \emptyset)}_{\text{no parents}} + \underbrace{s(C, \{A, B\})}_{\text{parents } A, B} + \underbrace{s(D, \{C\})}_{\text{parent } C}$$

Practical consequence

When we *add* $A \rightarrow C$ to the graph, only the local score $s(C, \text{Pa}_C)$ changes.

We do **not** need to recompute the whole graph score — just one term.

Search Strategy I: Dynamic Programming (DP / EXACT)

Key idea: Exploit score decomposability to find the *globally optimal* DAG.

Optimal parent set:

$$\hat{\Pi}_i = \arg \max_{\Pi \subseteq V \setminus \{i\}} s_i(X_i, \Pi)$$

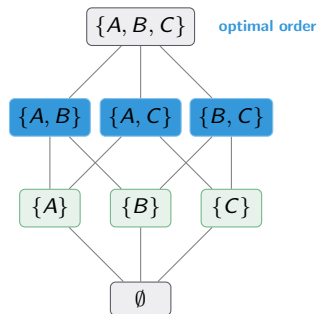
Precompute for all i and all subsets Π . (*Exponential in n but tractable for small n*)

Optimal ordering:

$$\hat{\sigma} = \arg \max_{\sigma \in \text{Orders}} \sum_i s_i(X_i, \hat{\Pi}_i^{\sigma})$$

where $\hat{\Pi}_i^{\sigma}$ is the best parent set *respecting order σ* .

DP over the lattice of variable subsets solves this in $O(n^2 \cdot 2^n)$ instead of super-exponential brute force.



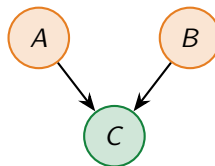
DP over subsets $S \subseteq V$, $|V| = n$

Dynamic programming example

Three variables: $\{A, B, C\}$. We compute local scores $s_i(\text{var}, \text{parents})$ from data.

Variable	Parents	Score
A	$\emptyset$	-8
A	$\{B\}$	-6
B	$\emptyset$	-7
B	$\{A\}$	-9
C	$\{A, B\}$	-5
C	$\{B\}$	-11
C	$\{A\}$	-10

Best DAG found by DP



Total score: $-8 + (-7) + (-5) = -20$
 (using $s(A, \emptyset) + s(B, \emptyset) + s(C, \{A, B\})$)

DP step: For each ordering (e.g. $A \prec B \prec C$), assign each variable its best parents from preceding variables.
 Pick the ordering with the highest total score. For $n = 3$: only $3! = 6$ orderings.

Search strategy II: Greedy Equivalence Search (GES)

Key idea: Search over the space of *CPDAGs* (equivalence classes) in two phases.

❶ **Forward phase (FES):**

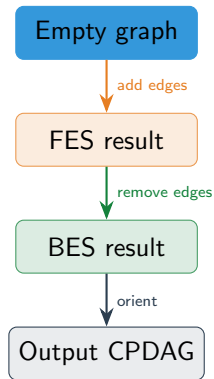
Start from empty graph. Greedily add the edge that most *increases* the score.

Stop when no single-edge addition improves score.

❷ **Backward phase (BES):**

Greedy remove the edge that most *increases* the score.

Stop when no single-edge removal improves score.



Consistency

GES is **consistent** under the Faithfulness assumption:
with enough data it recovers the true CPDAG.

Other algorithms

- Exact
 - Integer linear programming (ILP)
 - Max-SAT
 - A^* search
- Heuristics
 - Greedy/local search
 - Ordering-based
 - Continuous relaxations
- Hybrid strategies
 - Min-max hill-climbing

3. Learning beyond Markov equivalence classes

Learning beyond Markov equivalence classes

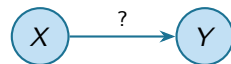
- So far, we have shown that given **causal Markov condition**, **causal sufficiency**, **faithfulness** and **conditional independence oracle** it is possible to learn the causal model up to Markov equivalence class from **observational data**
- What if we want to learn the actual graph?
- Solution 1: **Use experimental data**
- Solution 2: **Make more assumptions**
- Solution 3: Use domain knowledge

3.1. Learning from interventional data

From observation to intervention

The observational limit:

- Structure learning from observational data returns a **Markov Equivalence Class (MEC)**
- All DAGs in a MEC share the same skeleton and v-structures
- Edge *directions* within the MEC are unidentifiable from observations alone



observ.

$X \rightarrow Y$ and $Y \rightarrow X$
are indistinguishable



do(X)

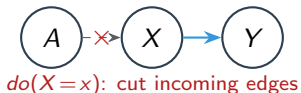
If Y changes:
 $X \rightarrow Y$ confirmed

Interventions help by:

- Breaking distributional symmetry between MECs
- Directly testing the direction of causal influence

How to direct edges given interventional data

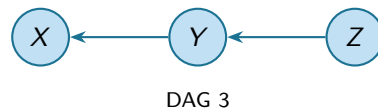
- Key idea: If intervening X changes the distribution of Y but intervening Y does not change the distribution of X then $X \rightarrow Y$



Example

Setup: Variables X, Y, Z . Observational MEC contains two DAGs.

Observational MEC



Intervention on Y : $\text{do}(Y = y)$

- **DAG 1:** $P(X|\text{do}(Y)) = P(X)$,
 $P(Z|\text{do}(Y)) \neq P(Z)$
- **DAG 2:** $P(X|\text{do}(Y)) \neq P(X)$,
 $P(Z|\text{do}(Y)) \neq P(Z)$
- **DAG 3:** $P(X|\text{do}(Y)) \neq P(X)$,
 $P(Z|\text{do}(Y)) = P(Z)$

Conclusion: $\text{do}(X)$ distinguishes the Markov equivalent DAGs!

Challenges with interventional data

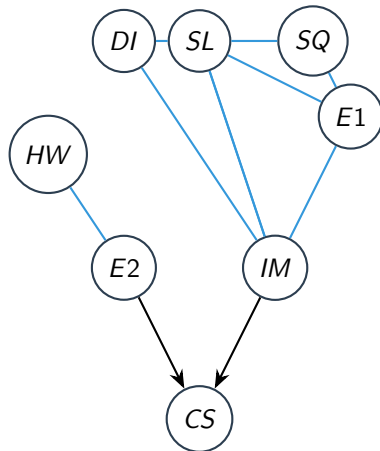
- **Cost:** Interventions (experiments) may be expensive or harmful
- **Unknown targets:** Not always known which variable was actually perturbed
- **Off-target effects:** Intervention may affect multiple variables
- **Noise:** Limited samples per condition

Exercise

Discuss in groups. Which variable would you intervene? Why?

- SL = Stress level
- SQ = Sleep quality
- E1 = Exercise
- DI = Diet
- HW = Handwashing
- E2 = Exposure
- IM = Immune strength
- CS = Cold severity

Observational MEC



3.2. Learning with functional assumptions

Learning with functional assumptions

- So far, we have not made any assumptions about functions $\mathbf{F}$ or distributions $P(\mathbf{U})$.
- However, stronger assumptions may help us to identify more causal relations

Two-variable identifiability problem

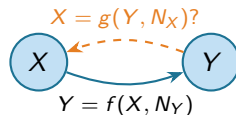
Fundamental question:

Given joint data on (X, Y) , can we determine whether $X \rightarrow Y$ or $Y \rightarrow X$?

Without additional assumptions: **No.**

For any distribution $P(X, Y)$, there exists a model $X \rightarrow Y$ and a model $Y \rightarrow X$ consistent with it.

Functional assumptions restrict the model class to break this symmetry.



**Both consistent
with $P(X, Y)$ in general**

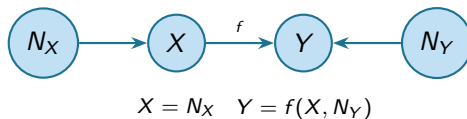
Structural Causal Models (SCMs) - Refresher

Definition

An SCM assigns to each variable X_i a function: $X_i = f_i(\text{Pa}(X_i), N_i)$ where N_i are jointly independent noise variables.

Key ingredients:

- DAG $\mathcal{G}$ (structure)
- Functions f_i (mechanism)
- Noise distributions $P(N_i)$



Identifiability question:

Given $P(X, Y)$, under what restrictions on f and $P(N)$ is the direction $X \rightarrow Y$ vs $Y \rightarrow X$ uniquely determined?

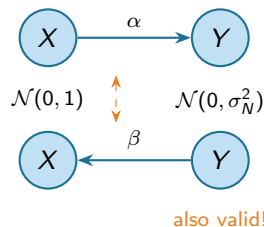
Linear Gaussian: *Not* identifiable

Model: $Y = \alpha X + N_Y$, $X \sim \mathcal{N}(0, \sigma_X^2)$,
 $N_Y \sim \mathcal{N}(0, \sigma_N^2)$

Result: A linear Gaussian model $X \rightarrow Y$ can
 always be *reversed* to a valid $Y \rightarrow X$ model.
 Direction is **not** identified.

Why? The joint Gaussian distribution $P(X, Y)$
 is symmetric with respect to conditional
 factorizations:

$$P(X, Y) = P(X) P(Y|X) = P(Y) P(X|Y)$$



Both models fit the same bivariate Gaussian

Linear Non-Gaussian: LiNGAM

Model: $Y = \alpha X + N_Y$, N_Y non-Gaussian

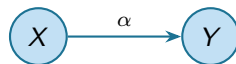
Key insight:

If $X \perp\!\!\!\perp (Y - \hat{\alpha}X)$, then $X \rightarrow Y$ is the correct direction.

In the reverse model $X = \beta Y + \tilde{N}_X$, the residual $\tilde{N}_X$ is *dependent* on Y (unless N_Y is Gaussian).

Consequence: Under this setting, direction between to variables can be detected.

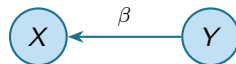
Direction identifiable:



N_Y non-Gaussian

Residual $Y - \hat{\alpha}X \perp X$
only for true direction

Wrong direction:



Residual $\tilde{N}_X \not\perp Y$
(dependence detected)

Learning with functional assumptions

- There exists other types of models that are identifiable (e.g., additive noise models)
- Bottom line: Sometimes it is possible to identify the causal direction between two variables with (strong) functional assumptions
 - Check your data whether your assumptions are reasonable

4. Evaluation

G.E. Box

All models are wrong but some are useful.

Challenges in evaluation

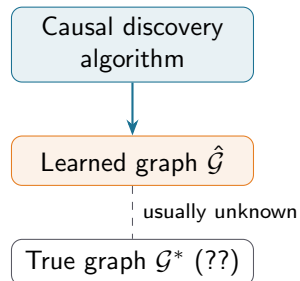
With ground truth: Compare learned graph to true graph using SHD, precision/recall, SID, ...

Without ground truth (real data):

- The “true” causal structure is unknown
- Benchmark datasets are scarce
- Evaluation must rely on *proxies*

Approaches available:

- 1 Predictive / statistical fit
- 2 Interventional validation
- 3 Expert knowledge comparison
- 4 Literature-based benchmarks
- 5 Simulation with realistic data



Metrics when ground truth is known

Metric	Description	Unit
SHD	Structural Hamming Distance	# edge ops
Precision	True directed edges / learned directed	rate
Recall	True directed edges / true directed	rate
F1	Harmonic mean of P and R	rate
SID	Structural Intervention Distance	count
Ancestor precision/recall	Correct ancestral relations	rate

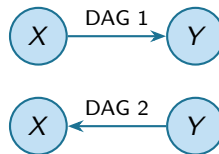
SID (Peters & Bühlmann 2015)

Counts the number of interventional distributions that are *wrongly estimated* by the learned graph. More relevant for causal inference tasks than SHD.

Approach 1: Predictive evaluation

Idea: Use the learned DAG to predict held-out data.

- Fit a DAG model to training data; evaluate likelihood or RMSE on test data
- Compares Markov factorizations, not causal directions



Same predictive likelihood!
(same MEC)

Pros:

- Fully automated, no expert needed
- Standard ML workflow

Cons:

- Many DAGs in the MEC have *identical* predictive performance
- Does not distinguish causal from spurious edges
- A reversed edge may score equally well

Approach 2: Held-out interventional data

Idea: Reserve some interventional data for evaluation.

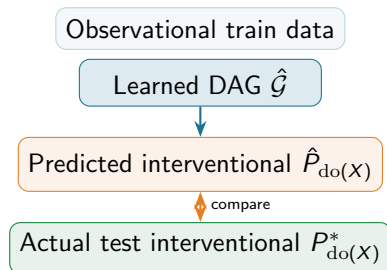
- 1 Learn causal graph on observational (or training) data
- 2 Use test interventional data: check predicted vs. actual post-intervention distributions

Pros:

- Genuinely tests causal structure (not just MEC)
- Principled and quantitative

Cons:

- Requires interventional test data – often costly
- Interventions may target only a subset of variables



Approach 3: Expert knowledge and literature

Idea: Compare learned edges to known causal relations from domain experts or published studies.

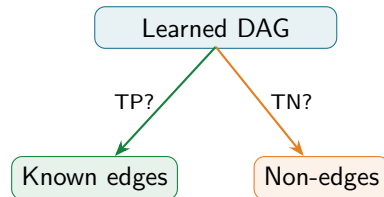
- **Positive controls:** known causal pairs should be recovered
- **Negative controls:** known non-causal pairs should be absent

Pros:

- Directly relevant to domain
- Can validate directions, not just skeleton

Cons:

- Expert knowledge may be incomplete or wrong
- Selection bias: only well-studied edges can serve as controls
- Partial: cannot assess recall of unknown edges



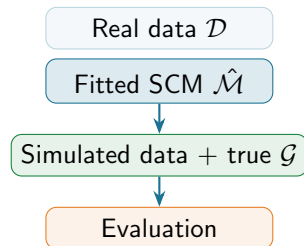
Approach 4: Simulations

Idea: Use real data to fit a model, then *simulate* from the model.

- 1 Fit structural model to real data (estimate DAG, mechanisms, noise)
- 2 Generate synthetic datasets from the fitted model
- 3 Compare data sets to real data

Idea: Sensitivity analysis.

- 1 Perturb the data (or model/algorithm hyperparameters)
- 2 Fit structural model to the perturbed data (estimate DAG, mechanisms, noise)
- 3 Compare different models and assess stability.

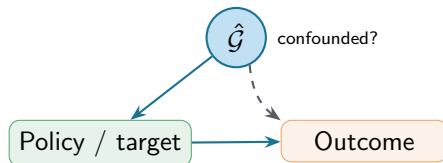


Approach 5: Downstream task evaluation

Idea: Use the learned causal model for a downstream task; evaluate task performance.

• **Examples:**

- Drug target identification: does the predicted target lead to the right therapeutic effect?
- Policy evaluation: does the estimated causal effect match a natural experiment?
- Feature selection: do causal parents improve prediction under distribution shift?



Pros:

- Directly measures practical value

Cons:

- Confounds graph quality with model quality
- May not distinguish causal from predictive accuracy

Comparison of evaluation strategies

Strategy	Tests direction?	Needs int. data?	Cost	Reliability
Predictive fit	No	No	Low	Low
Held-out interventions	Yes	Yes	High	High
Expert knowledge	Partial	No	Medium	Medium
Simulations	No	No	Low	Medium
Downstream task	Partial	Sometimes	Varies	Varies

Recommendation

Use **multiple complementary strategies**. No single approach is sufficient. Predictive fit should not be the sole criterion: two DAGs in the same MEC score identically but may have very different causal interpretations.

Executive Summary

- Lack of accurate models is a bottleneck in causal modelling
- Learning causal models from data is possible
 - to some degree
 - with strong assumptions